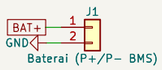
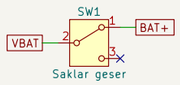
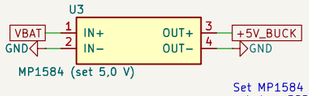
1. DAYA (wiring diagram blok 1)

J1 = P+/P− BMS (baterai 2S di rangka)





SW1: kaki tengah (B) = VBAT

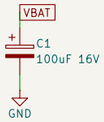


Set MP1584 ke 5,0 V dulu sebelum ESP32 dipasang

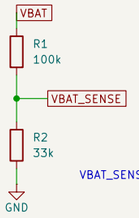
D1: cegah arus balik saat USB dicolok



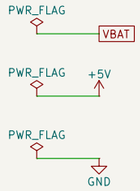
PWR_FLAG: daya VBAT, +5V, GND datang dari konektor/dioda



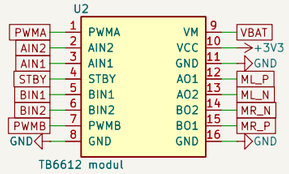
C1 dekat pin VM TB6612

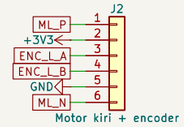


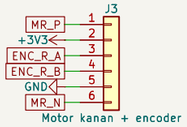
VBAT_SENSE ke GPIO39 (ADC1)



2. DRIVER MOTOR + MOTOR (wiring diagram blok 2)







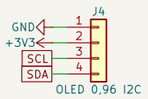
Urutan pin modul TB6612 beda tiap penjual: cocokkan dengan modul yang dibeli.

VM = VBAT (7,4 V) langsung, VCC = 3,3 V logika. AO ke motor kiri, BO ke motor kanan.

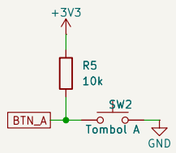
J2/J3 = JST PH 6 pin. Encoder tanpa resistor (firmware INPUT_PULLUP).

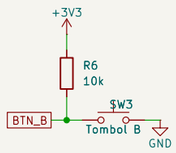
Urutan 6 kabel motor N20 beda tiap penjual: cek sebelum crimp.

3. LAYAR, TOMBOL, BUZZER (wiring diagram blok 3)

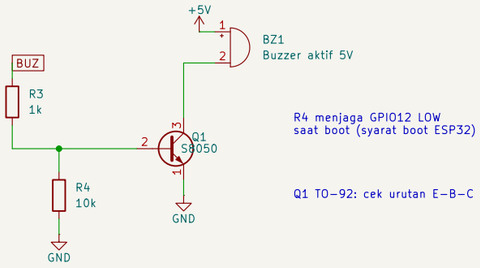


Cek urutan pin modul OLED (ada yang VCC dulu baru GND)





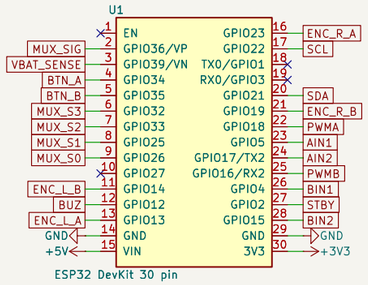
R5/R6 wajib: GPIO34/35 tanpa pull-up internal



R4 menjaga GPIO12 LOW saat boot (syarat boot ESP32)

Q1 TO-92: cek urutan E-B-C

4. ESP32 DEVKIT (otak)



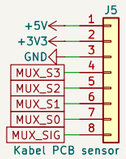
GPIO34/35/36/39: hanya input

GPIO2 (STBY) dan GPIO12 (BUZ) harus LOW saat boot

EN, GPIO27, TX0, RX0: tidak dipakai (X)

Footprint W25.4: ukur jarak baris pin modul (22,86 mm -> W22.86)

5. KABEL KE PCB SENSOR (wiring diagram blok 4)



J5 = JST XH 8 pin, urutan pin SAMA dengan J1 PCB sensor (kabel lurus, pin n ke pin n)

Ikuti NOMOR pin: pin 4–7 terbalik dari urutan nama (S3 dulu)

Di PCB: J5 sejajar tepi miring sidepod kiri, pin 1 di ujung dekat hidung

Dibuat oleh hardware/tools/gen_sch_utama.py, pin ESP32 sesuai CLAUDE.md
Panduan per blok: docs/wiring-diagram.html
P01 Kel.5 PjBL DTEO FV ITS

Sheet: /		
File: pcb-utama.kicad_sch		
Title: LTM – PCB utama		
Size: A3	Date: 2026-10-22	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1